

Arcana Match: Balance & Economy

Design workbook by Umut Eren Demirci. Blue cells are inputs; everything else is calculated.

Sheet	What it contains	Live formulas?
Level Curve	Goals, moves, pressure and card reward for levels 1-40, driven by the parameters at the top (mirrors LevelConfig.cs).	Yes
Spawn Odds	Tile odds for one- and two-goal levels as a function of the goal-suit boost (mirrors SpawnWeights.cs).	Yes
Simulation	Bot results for levels 1-30 with and without the 1.4x goal boost (LevelSimulator, 60 games per cell).	Summary rows
Boost Sweep	Win rates for boosts 1.0x-1.7x, used to choose 1.4x; counts levels inside the target band.	Yes
Specials	The five special tiles: how they are made, what they clear, and their priority.	Expected clears
Scoring & Combo	Points per tile and chime pitch for each cascade step; an example cascade.	Yes
Boosters & Economy	Starting boosters, refills, resource sources and sinks, and proposed monetisation.	No
Tuning	Animation and feel values from the GameTuning asset.	No
Audio	Every sound effect, its trigger, mix level and design intent.	No

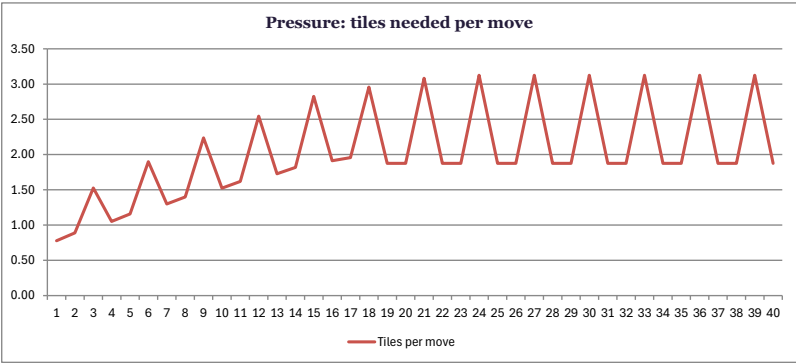
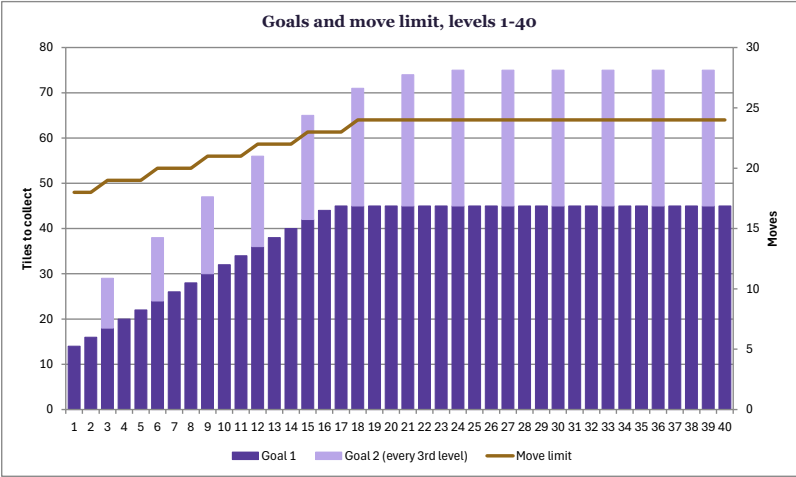
Source of truth: the game's code (Assets/_Project/Scripts/Core). Simulation data comes from running the shipped rules with a bot (Arcana Match > Debug > Run Balance Simulation). Prices on the economy sheet are proposals, not tested numbers.

Level curve

Change the blue parameters and every level updates. Mirrors LevelConfig.For(n).

Base goal (tiles)	12	Suit rotation
Goal growth per level	2	Wands
Goal cap	45	Cups
Second goal base	8	Swords
Second goal cap	30	Pentacles
Second goal every N levels	3	
Base moves	18	
+1 move every N levels	3	
Max extra moves	6	
Foil: moves left to earn it	5	

Level	Goal 1	Goal 1 suit	Goal 2	Goal 2 suit	Total tiles	Moves	Tiles per move	Card reward	Beat	Foil needs goals in
1	14	Wands			14	18	0.78	The Fool		13 moves
2	16	Cups			16	18	0.89	Random Cups card		13 moves
3	18	Swords	11	Pentacles	29	19	1.53	Random Swords card	Spike	14 moves
4	20	Pentacles			20	19	1.05	Random major arcana	Breather	14 moves
5	22	Wands			22	19	1.16	Random Wands card		14 moves
6	24	Cups	14	Swords	38	20	1.90	Random Cups card	Spike	15 moves
7	26	Swords			26	20	1.30	Random Swords card	Breather	15 moves
8	28	Pentacles			28	20	1.40	Random major arcana		15 moves
9	30	Wands	17	Cups	47	21	2.24	Random Wands card	Spike	16 moves
10	32	Cups			32	21	1.52	Random Cups card	Breather	16 moves
11	34	Swords			34	21	1.62	Random Swords card		16 moves
12	36	Pentacles	20	Wands	56	22	2.55	Random major arcana	Spike	17 moves
13	38	Wands			38	22	1.73	Random Wands card	Breather	17 moves
14	40	Cups			40	22	1.82	Random Cups card		17 moves
15	42	Swords	23	Pentacles	65	23	2.83	Random Swords card	Spike	18 moves
16	44	Pentacles			44	23	1.91	Random major arcana	Breather	18 moves
17	45	Wands			45	23	1.96	Random Wands card		18 moves
18	45	Cups	26	Swords	71	24	2.96	Random Cups card	Spike	19 moves
19	45	Swords			45	24	1.88	Random Swords card	Breather	19 moves
20	45	Pentacles			45	24	1.88	Random major arcana		19 moves
21	45	Wands	29	Cups	74	24	3.08	Random Wands card	Spike	19 moves
22	45	Cups			45	24	1.88	Random Cups card	Breather	19 moves
23	45	Swords			45	24	1.88	Random Swords card		19 moves
24	45	Pentacles	30	Wands	75	24	3.13	Random major arcana	Spike	19 moves
25	45	Wands			45	24	1.88	Random Wands card	Breather	19 moves
26	45	Cups			45	24	1.88	Random Cups card		19 moves
27	45	Swords	30	Pentacles	75	24	3.13	Random Swords card	Spike	19 moves
28	45	Pentacles			45	24	1.88	Random major arcana	Breather	19 moves
29	45	Wands			45	24	1.88	Random Wands card		19 moves
30	45	Cups	30	Swords	75	24	3.13	Random Cups card	Spike	19 moves
31	45	Swords			45	24	1.88	Random Swords card	Breather	19 moves
32	45	Pentacles			45	24	1.88	Random major arcana		19 moves
33	45	Wands	30	Cups	75	24	3.13	Random Wands card	Spike	19 moves
34	45	Cups			45	24	1.88	Random Cups card	Breather	19 moves
35	45	Swords			45	24	1.88	Random Swords card		19 moves
36	45	Pentacles	30	Wands	75	24	3.13	Random major arcana	Spike	19 moves
37	45	Wands			45	24	1.88	Random Wands card	Breather	19 moves
38	45	Cups			45	24	1.88	Random Cups card		19 moves
39	45	Swords	30	Pentacles	75	24	3.13	Random Swords card	Spike	19 moves
40	45	Pentacles			45	24	1.88	Random major arcana	Breather	19 moves



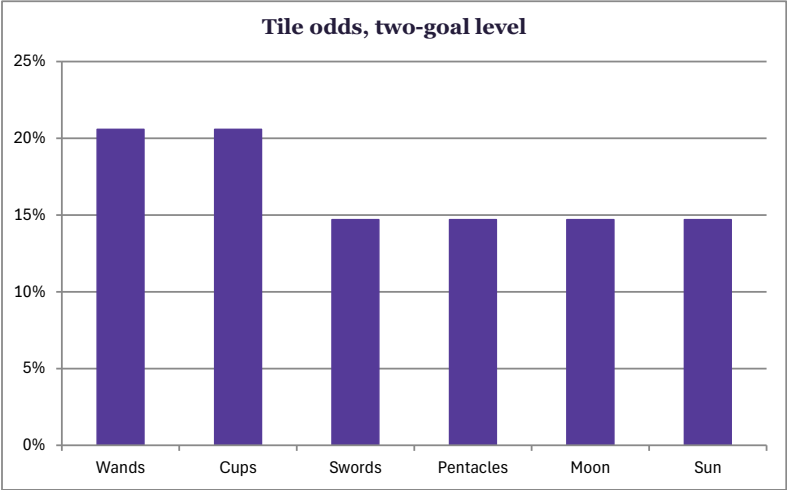
Goal-tile spawn odds

Goal suits get extra weight on the starting board and in refills. Mirrors SpawnWeights.cs and LevelConfig.GoalSpawnBoost.

Goal-suit weight (boost)	1.4
Tiles on the board	64
Tiles refilled per move (typical)	7

Tile	Kind	One-goal level (Wands)			Two-goal level (Wands + Cups)		
		Is goal	Weight	Chance	Is goal	Weight	Chance
Wands	Suit	1	1.4	21.9%	1	1.4	20.6%
Cups	Suit	0	1	15.6%	1	1.4	20.6%
Swords	Suit	0	1	15.6%	0	1	14.7%
Pentacles	Suit	0	1	15.6%	0	1	14.7%
Moon	Filler	0	1	15.6%	0	1	14.7%
Sun	Filler	0	1	15.6%	0	1	14.7%
Chance a new tile is a goal suit		21.9%			41.2%		
Without the boost		16.7%			33.3%		
Expected goal tiles on a fresh board		14.0			26.4		
Expected goal tiles per refill		1.5			2.9		

Why 1.4x: it raises a single goal suit from 16.7% to 21.9% of new tiles. The boost sweep sheet shows that this lifts late-level win rates for a goal-focused bot into the 75-95% target band without making early levels trivial or the board feel rigged.

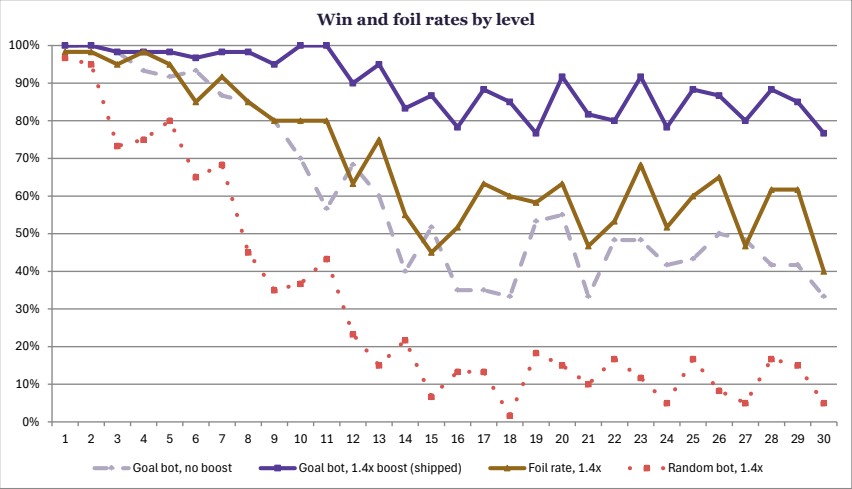


Balance simulation, levels 1-30

LevelSimulator plays the shipped rules. Goal-focused bot = picks the move that clears the most goal tiles. 60 games per cell.

Level	Moves	Goal 1	Suit	Goal 2	Suit	Goal bot, no boost	Goal bot, 1.4x boost	Foil rate, 1.4x	Random bot, 1.4x	Avg moves left on win	Boost gain
	1	18	14 Wands			100%	100%	98%	97%	11.9	0%
	2	18	16 Cups			100%	100%	98%	95%	11.6	0%
	3	19	18 Swords		11 Pentacles	98%	98%	95%	73%	10.4	0%
	4	19	20 Pentacles			93%	98%	98%	75%	11.0	+5%
	5	19	22 Wands			92%	98%	95%	80%	10.3	+7%
	6	20	24 Cups		14 Swords	93%	97%	85%	65%	9.5	+3%
	7	20	26 Swords			87%	98%	92%	68%	10.0	+12%
	8	20	28 Pentacles			85%	98%	85%	45%	8.5	+13%
	9	21	30 Wands		17 Cups	80%	95%	80%	35%	8.7	+15%
	10	21	32 Cups			70%	100%	80%	37%	8.0	+30%
	11	21	34 Swords			57%	100%	80%	43%	8.4	+43%
	12	22	36 Pentacles		20 Wands	68%	90%	63%	23%	6.4	+22%
	13	22	38 Wands			60%	95%	75%	15%	8.3	+35%
	14	22	40 Cups			40%	83%	55%	22%	6.7	+43%
	15	23	42 Swords		23 Pentacles	52%	87%	45%	7%	6.0	+35%
	16	23	44 Pentacles			35%	78%	52%	13%	6.4	+43%
	17	23	45 Wands			35%	88%	63%	13%	6.5	+53%
	18	24	45 Cups		26 Swords	33%	85%	60%	2%	6.4	+52%
	19	24	45 Swords			53%	77%	58%	18%	7.7	+23%
	20	24	45 Pentacles			55%	92%	63%	15%	6.7	+37%
	21	24	45 Wands		29 Cups	33%	82%	47%	10%	5.5	+48%
	22	24	45 Cups			48%	80%	53%	17%	6.5	+32%
	23	24	45 Swords			48%	92%	68%	12%	7.1	+43%
	24	24	45 Pentacles		30 Wands	42%	78%	52%	5%	6.4	+37%
	25	24	45 Wands			43%	88%	60%	17%	6.4	+45%
	26	24	45 Cups			50%	87%	65%	8%	7.1	+37%
	27	24	45 Swords		30 Pentacles	48%	80%	47%	5%	5.6	+32%
	28	24	45 Pentacles			42%	88%	62%	17%	7.2	+47%
	29	24	45 Wands			42%	85%	62%	15%	6.7	+43%
	30	24	45 Cups		30 Swords	33%	77%	40%	5%	5.2	+43%
Levels 1-10						90%	98%	91%	67%	10.0	8%
Levels 11-20						49%	88%	61%	17%	7.0	39%
Levels 21-30						43%	84%	56%	11%	6.4	41%
Levels below 75% (goal bot)						21	0				

Reading it: without the boost the goal-focused bot fell to 33-42% on late levels (big goals outgrew what the board delivered). With 1.4x it stays at 77-100%, foil stays clearly harder than a win, and random play fails most late levels, so decisions matter.



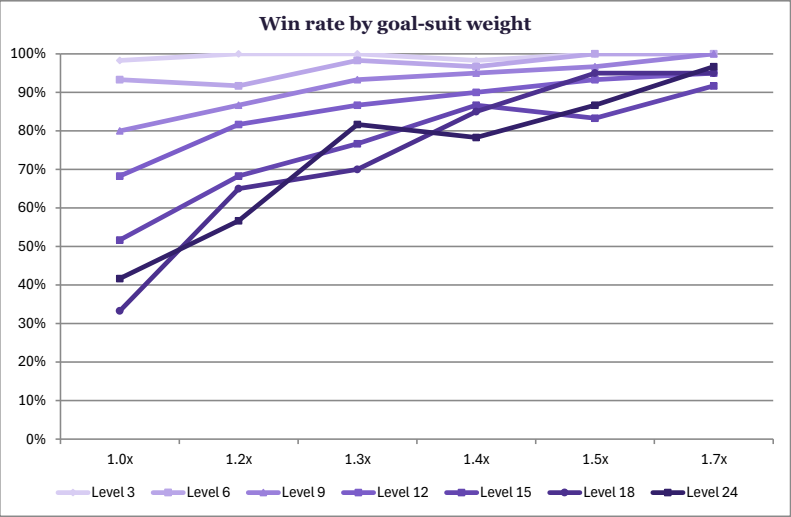
Goal-boost sweep

Win rate of the goal-focused bot by goal-suit weight. Target band for a skilled player: 75-95%.

Target band low	75%
Target band high	95%

Boost	Level 3	Level 6	Level 9	Level 12	Level 15	Level 18	Level 24	Levels in band	Lowest	Average
1.0x	98%	93%	80%	68%	52%	33%	42%	2	33%	67%
1.2x	100%	92%	87%	82%	68%	65%	57%	3	57%	79%
1.3x	100%	98%	93%	87%	77%	70%	82%	4	70%	87%
1.4x	98%	97%	95%	90%	87%	85%	78%	5	78%	90% <- shipped
1.5x	100%	100%	97%	93%	83%	95%	87%	4	83%	94%
1.7x	100%	100%	100%	95%	92%	95%	97%	3	92%	97%

Decision: 1.4x puts the most tested levels inside the target band while keeping the lowest level near 78%. Above 1.5x, early and mid levels pass 95% and goals stop shaping decisions.



Special tiles

How each special is made and what it clears on the 8 x 8 board. Priority decides overlaps: the rarer special wins.

Priority	Special	Made by	Pattern	Effect	Tiles cleared (8 x 8)	Chains?	Visual	Sound
1	Arcana tile	Straight line of 5+	XXXXX	Swap with a tile: clears every tile of that type (x2 score). Two Arcana: whole board. Hit by an effect: clears the most common type.	10.7	Yes	Spinning rainbow with an eye	Electric crackle / rush and boom
2	Star	L or T shape, 5+ tiles	XXX / X. . / X. .	Clears its whole row and column	15.0	Yes	Four-point star of light	Bright zing, pitched up
3	Column clearer	Horizontal line of 4	XXXX	Clears its column	8.0	Yes	Vertical stripes	Zing
3	Row clearer	Vertical line of 4	X / X / X / X	Clears its row	8.0	Yes	Horizontal stripes	Zing
4	Wheel	2 x 2 square	XX / XX	Clears the 3 x 3 block around it	9.0	Yes	Slowly turning ring of beads	Deep thump + small shake
	Tower Strike (booster)	Booster, tap a tile		Clears the 3 x 3 block around the tapped tile, firing specials	9.0	Starts chains	Lightning bolt, flash, shake	Thunder

Arcana expected clear assumes an even mix of six tile types (64 / 6). Placement: a special made by the player's swap appears on the swapped cell; a cascade-made special appears in the middle of its shape (the corner of an L). Each special fires once per chain, so chains always end.

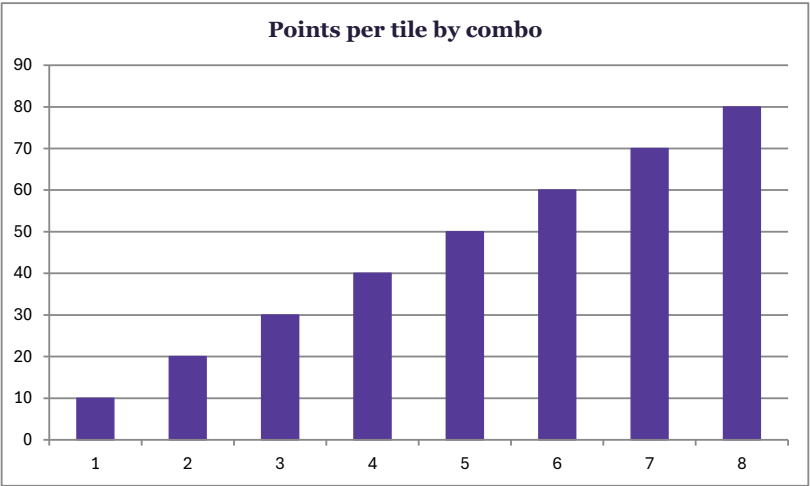
Scoring and combo

Points = tiles cleared x points per tile x combo. Combo rises by 1 with each cascade; the match chime climbs a pentatonic scale.

Points per tile 10

Combo	Interval	Chime pitch	Points per tile	Example: tiles cleared	Points this step	Running total
1	Root	1.000x	10	3	30	30
2	Major 2nd	1.125x	20	3	60	90
3	Major 3rd	1.250x	30	4	120	210
4	Perfect 5th	1.500x	40	3	120	330
5	Major 6th	1.667x	50	5	250	580
6	Octave	2.000x	60	3	180	760
7	Octave (held)	2.000x	70	3	210	970
8	Octave (held)	2.000x	80	3	240	1210

Pentatonic steps never clash, so however long a chain runs the chimes stay musical. The same combo number multiplies score, so a cascade is both heard and rewarded. Pitches live in the SoundBank asset (comboPitches).



Boosters and economy

All purchases and ads go through IStoreService / IAdService. The build uses a placeholder that grants everything for free.

Boosters (live)

Booster	Start	Refill (+)	Effect	Costs a move?	Design role
Insight	3	3	An eye opens and two orbs mark a valid move	No	Removes frustration when stuck; teaches patterns
Tower Strike	2	3	Armed first; tap a tile to clear its 3 x 3 with lightning, firing specials	No	Finish a goal or break a stubborn corner
Reshuffle	2	3	Gather the tiles and deal them again	No	A fresh board when the layout looks hopeless

Resources: sources and sinks (live)

Resource	Source	Sink	Notes
Moves	Level move limit (18-24)	1 per accepted swap	Invalid swaps are free
Extra moves (+5)	Lose screen, once per attempt	Continues the level	Free in this build
Boosters	Starting stock; +3 per refill	1 per use	Never cost a move
Cards	1 per level won	None	Never spent or lost
Foil	Win with 5+ moves left	None	Permanent; never downgraded

Monetisation (proposed, not active)

Offer	Where	Rewarded ad	Purchase	Proposed price (USD)	Guardrail
Extra moves	Lose screen	Watch for +5 moves	+5 moves		\$0.99 Once per attempt; never a paywall loop
Booster refill	Tap an empty booster	Watch for +1	3 of one booster		\$1.99 Boosters are optional
Booster pack	Refill screen	No	3 of each booster		\$4.99 Saves time only
Card sleeves	Collection	No	Cosmetic card backs		\$2.99 Cards themselves are never sold
Remove ads	Settings	n/a	One-time		\$3.99 Rewarded ads stay opt-in

Guardrails: no energy or lives system, no card can be bought, ads are always opt-in and rewarded, no interstitials mid-level. Prices are placeholders to validate in a soft launch.

Feel tuning

Values from Assets/_Project/Data/GameTuning.asset. Taken from the browser prototype; editable during Play mode.

Group	Field	Value	Unit	What it does
Movement	moveDuration		0.26 s	Tile glide time for swaps and falls
Movement	moveBezier	(.25, .8, .35, 1)	curve	CSS cubic-bezier easing from the prototype
Swapping	swapHold		0.21 s	Beat after a swap before clearing
Swapping	invalidShakeDuration		0.25 s	Head-shake on an invalid swap
Swapping	invalidShakeAmount		0.07 tiles	Shake distance
Swapping	swipeThreshold		0.35 tiles	Drag distance that counts as a swipe
Clearing	popDuration		0.2 s	Tiles grow and fade
Clearing	popScale		1.35 x	How big a clearing tile grows
Clearing	clearHold		0.22 s	Beat after a clear
Falling	fallHold		0.3 s	Beat after gravity and refill
Falling	spawnDuration		0.3 s	New tiles drop in
Selection	selectedScale		1.12 x	Selected tile size
Selection	hintPeriod		0.9 s	Hint pulse period
Selection	hintScale		1.16 x	Hint pulse size
Particles	particlesPerTile		5 count	Sparks per cleared tile (small clears)
Particles	particlesPerTileMedium		3 count	From 10 tiles cleared
Particles	particlesPerTileLarge		2 count	From 24 tiles cleared, to keep big blasts readable
Floater	floaterDuration		0.9 s	Score floater life
Floater	floaterRise		0.55 tiles	How far it rises
Reshuffle	gatherDuration		0.36 s	Tiles spin into the centre
Reshuffle	gatherHold		0.38 s	Pause in the centre
Reshuffle	dealDuration		0.42 s	Tiles dealt back out
Reshuffle	dealStagger		0.009 s	Delay between dealt tiles
Rules	Foil threshold		5 moves	Moves left to earn foil (LevelSession)
Rules	Extra moves		5 moves	Rescue amount, once per attempt (LevelSession)
Rules	Daily draw unlock		3 cards	Cards owned before the daily draw opens (DailyDraw)

Audio list

Every sound is synthesized by SoundGenerator.cs; mix levels live in the SoundBank asset.

Sound	Trigger	Mix	Description	Intent
Tap	Any button		0.35 Tiny, soft glassy click	Confirms input without noise
Swap	Tiles swap		0.45 Short airy whoosh that brightens as tiles slide	Motion
Invalid	Swap with no match		0.55 Two soft wooden bonks	A gentle 'nope', never a buzzer
Match	Tiles clear		0.60 FM glass bell; pitch follows the combo	Chains audibly build
Special	Special tile created		0.70 Quick rising arpeggio with a shimmer	You made something
Beam	Line or Star fires		0.60 Bright zing sweeping upward	Sells the light sweep
Arcana	Arcana tile fires		0.75 Electric crackle over a low hum	The biggest single moment
ArcanaBig	Two Arcana swapped		0.85 Swelling rush into a chord and a deep boom	The board-clear payoff
Strike	Tower Strike, Wheel (pitched down)		0.90 Sharp crack, rolling thunder, low boom	Weight and power
Insight	Insight booster		0.65 Ascending pentatonic sparkle with echo	Magical help
Shuffle	Reshuffle		0.60 Card riffle speeding up, four cards dealt	A real deck
Flip	Card turns over		0.60 Papery swish and soft landing	The reveal
Foil	Foil card shown		0.50 Scattered high sparkles	Holographic shine
Win	Level won		0.75 Rising arpeggio blooming into a chord	Celebration
Lose	Out of moves		0.60 Three soft falling notes	Disappointed but gentle
GoalDone	A goal is completed		0.60 Bright two-note ding-ding	Progress marker
Reward	Refill or extra moves granted		0.60 Quick coin-like chime	Something gained
Music: Night Sky	Always (mutable)		32 s seamless ambient loop, Dm - Bb - Gm - A, sparse bells and echo	Mood; sits under the effects